

Summer Project Report

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Design of 3D printed Orthopedic scaffolds

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1. INTRODUCTION

In the field of orthopedics, the treatment of diseases and damaged tissues has posed significant challenges. Various ailments, such as Osteoporosis, arthritis, and tissue destruction caused by trauma, have called for innovative solutions to aid in the regeneration and restoration of Orthopedic tissues. Techniques like autografting and allografting are the gold standard treatment in today's time. But it also has some drawbacks like the cell rejection in case of a foreign donor or morbidity in the tissues of patients from its donor site. As a result, tissue engineering emerged as a promising avenue for developing effective treatments, and 3D printed orthopedic scaffolds have gained considerable attention due to their potential to address these complex problems.

Traditional treatment methods, such as bone grafts or joint replacements, have limitations regarding availability, suitability, and long-term success rates.[1] Tissue engineering approach to regenerating damaged tissues by utilizing the body's innate regenerative capacity. The concept involves creating functional tissues or organ substitutes by combining cells, biomaterials, and biochemical factors in a controlled environment. One critical component of tissue engineering is using scaffolds, which serve as frameworks to guide and support the growth of new tissues. Among the various methods available for scaffold fabrication, 3D printing has emerged as a cutting-edge technology with tremendous potential. It allows for the precise and customizable fabrication of scaffolds with complex geometries and internal architectures that mimic the natural tissue structure. These 3D printed orthopedic scaffolds can be tailored to match the specific requirements of the defect site, such as size, shape, mechanical properties, and bioactivity, thereby enhancing their integration and promoting tissue regeneration. Current research in 3D printed orthopedic scaffolds focuses on refining the fabrication techniques, optimizing the scaffold design, and exploring new materials with improved biocompatibility and mechanical properties. Additionally, efforts are being made to improve the integration of scaffolds with the host tissue and to promote vascularization, which is vital for nutrient supply and waste removal in the newly formed tissues. By combining advances in biomaterials, fabrication techniques, and regenerative medicine, researchers aim to develop personalized and patient-specific solutions that can restore the functionality of damaged tissues and improve the quality of life for individuals affected by orthopedic diseases and injuries.

2. Literature survey

2.1. Bone Structure & Composition

Understanding the composition and structure of native bone tissue is necessary for successful materials design for bone-tissue engineering, as is choosing suitable biomimetic natural or synthetic materials, such as polymers, bioceramics, metals, and composites. Bone-tissue engineering has evolved from using biologically inert substances to bioactive materials that aim to regenerate functional bone tissue. Bone tissue is composed of cells, an extracellular matrix (ECM), and biologically active substances. The hierarchical structure of bone and its natural creation assist in bone-tissue engineering. The ECM, analogous to a bone-tissue-engineering scaffold, consists of crosslinked collagen fibrils and calcium phosphate minerals.[2][3] Carbonated hydroxyapatite, or bone mineral, is a significant component of bone tissue. The process of bioapatite formation in bones involves the deposition of calcium phosphate, neation, and crystallization into hydroxyapatite crystals. This process can be replicated on tissue engineered scaffolds to create biomimetic bone constructs with similar mechanical properties to natural tissue. Bone exhibits variable density, with dense cortical or compact bone on the outer regions and porous cancellous or spongy bone on the interior regions, consisting of trabeculae.[2] Materials such as polymers, ceramics, or their combinations are commonly used in bone-tissue engineering as membranes or three-dimensional scaffolds. These materials mimic the extracellular matrix (ECM) and provide mechanical support and a suitable environment for cell attachment, proliferation, and differentiation. Cells can be seeded onto the material or recruited from surrounding tissue after implantation. Although biological elements such as cells and growth hormones are valuable in bone-tissue creation, the price and regulatory restrictions might be prohibitive. As a result of this, materials-only aim to engage with the body's cells for bone restoration.

2.2. Bone Tissue Engineering :

Bone-tissue engineering research is to develop materials that can outperform traditional bone grafts and be remodeled by the patient's own cells. These materials are typically designed as scaffolds, providing a supportive structure for cellular attachment and the deposition of mineralized matrix. The design of these materials takes into account the properties of healthy bone tissue, such as its porous architecture, cellular infiltration, vascularization, and hierarchical organization. Clinical considerations play a significant role in materials design for bone-tissue engineering. Different types of bone defects, such as orthopedic or craniofacial defects, require

tailored strategies. Orthopedic defects occur in load-bearing bones like limbs and the spine, while craniofacial defects are generally non-load-bearing, affecting areas like the jaws or temporomandibular joints.[7] The causes of bone defects can range from injuries and diseases to congenital conditions or surgical removal of tumors. Age-related factors also need to be considered. For elderly individuals, the effects of natural aging on bone microstructure and regenerative potential must be taken into account. Bone-tissue engineering materials for pediatric patients need to accommodate ongoing growth and have dynamic or remodeling properties. There are also noteworthy sex-related differences in the microstructure and overall size of bones, which may influence materials design considerations.[4][5] Bone physiology involves the development and remodeling of bone tissue through processes like endochondral ossification and intramembranous ossification.[6] Osteoblasts play a key role in bone formation by secreting unmineralized collagen and promoting the deposition of calcium and phosphate minerals.[8] Osteoblasts eventually mature into osteocytes, which regulate bone homeostasis and respond to mechanical stimuli. Bone remodeling, driven by osteoclasts and osteoblasts, allows for functional adaptation, growth, and fracture healing. Macrophages and mesenchymal stem cells also contribute to bone healing and homeostasis.[9]

In biomaterials for bone-tissue engineering, bioceramics and polymers are often used to mimic the inorganic (hydroxyapatite) and organic (collagen) components of the bone extracellular matrix (ECM), respectively. Bioceramics can be crystalline bioactive ceramics or amorphous bioactive glasses, while polymers can be synthetic or derived from natural sources like collagen. Biodegradable metals and carbon-based nanomaterials are also utilized for their mechanical properties. Biomaterials can be designed to promote biochemical, mechanical, or piezoelectric stimulation of developmental signaling cascades, known as developmental engineering. Additionally, the nanointerfaces within bone can be mimicked using material nanotopography to direct macrophages towards bone-stimulating behavior.[10]

Understanding bone physiology allows for the biomimicry of candidate biomaterials in bone-tissue engineering, enabling the development of materials that replicate the structural and functional properties of bone tissue and promote its regeneration. Bone-tissue engineering systems can also be used to develop models for studying tumors, drug effects, and other bone-related diseases. Biocompatibility is a fundamental requirement for bone-tissue engineering biomaterials to ensure appropriate host response and avoid adverse outcomes. Surface characteristics that promote protein adsorption and cell binding are important for favorable interactions with the body. Compatibility with sterilization or disinfection techniques is also essential for eventual clinical translation. Bioactive materials can stimulate osteogenesis and promote bone formation, either at natural bone sites (orthotopic) or sites where bone tissue would not typically be present (ectopic).[3] Osteoinductive materials contain ions or growth factors that induce osteogenesis, while osteoconductive materials allow the deposition of

mineralized tissue on their surface and promote bonding with bone.[11] Mechanical and degradative properties of bone-tissue engineering materials should be tailored to facilitate tissue regeneration. Materials should withstand compressive loads and match the mechanical properties of native bone to prevent stress shielding, where the implant bears most of the load instead of the surrounding bone, leading to reduced bone density and weakening of the tissue.[8] In bone-tissue engineering, implanted biomaterial scaffolds provide mechanical support to growing bone tissue and undergo biodegradation.[13] The rate of scaffold degradation should be synchronized with the deposition of mineralized tissue to maintain the overall function of the system.[12] The degradation products must be biocompatible and not interfere with scaffold mineralization or alter the local pH.[14] The shape of the exterior surface of scaffolds should match the geometry of the defect site. The interior architecture often includes interconnected pores to facilitate vascularization and cell infiltration.[15][16] Pores with a diameter of 100-300 μ m are necessary for the ingrowth of blood vessels, nerve fibers, and osteoprogenitor cells.[17] Soft scaffolds made of adaptable hydrogels can enable cell migration without requiring pores or material degradation. Mechanical properties are critical for large, load-bearing defects, while minimally invasive methods may prefer weaker injectable hydrogels for small non-load-bearing defects.[18][19] The choice of materials and their properties can be tailored to meet specific clinical needs. High bioactivity and rapid degradation may be more valuable in some cases, as long as the regenerated tissue acquires the necessary mechanical properties for adequate tissue function within a suitable time frame.[20][21]

2.2.1. Materials for Scaffold Fabrication :

2.2.1.1. Polymers

Bone tissue engineering utilizes scaffolds and polymers to facilitate bone regeneration. Natural polymers such as collagen and gelatin are hydrophilic and can form hydrogels.[22][23] They possess desirable characteristics for bone-tissue engineering, including cell attachment ligands and enzymatic degradation properties. Alginate, derived from algae, can be modified to promote cell attachment and growth factor delivery.[25] Silk has outstanding mechanical properties and can guide hydroxyapatite formation.[24] Other natural polymers used include poly(hydroxyalkanoate), chitosan, gellan gum, and poly(phosphate).[26][27] Synthetic polymers offer more flexibility for chemical modifications and tailoring properties to specific applications. Poly(ϵ -caprolactone) (PCL) is known for its high mechanical strength and FDA-approved status, although it requires modifications to enhance cell adhesion.[28][29] Poly(propylene fumarate) exhibits bone-matching mechanical properties and biocompatible degradation products. Poly(L-lactic acid) (PLLA) and poly(lactic-co-glycolic acid) (PLGA) are degradable polymers

with tunable degradation rates. Poly(ethylene glycol) (PEG) is a highly hydrophilic polymer commonly used for hydrogel development, and poly(N-isopropylacrylamide) is a thermoresponsive polymer suitable for injectable hydrogels.[30][31]



Fig.1. PLA Filament roll

2.2.1.2. Bioceramics

Bioceramics are inorganic biomaterials used in bone-tissue engineering. They can be categorized as ceramics, glasses, or glass ceramics. Bioactive bioceramics can form a direct bond with surrounding bone tissue, while inert bioceramics cannot. However, research is focusing on the capacity of bioceramics to release biologically active ions for desired biological effects. Calcium phosphates, such as hydroxyapatite and β -tricalcium phosphate, are the most common bioactive ceramics used in bone-tissue engineering.[32] They mimic the inorganic phase of bone and can be used as injectable bone cements or coatings on implants. Hydroxyapatite exhibits slow degradation, while β -tricalcium phosphate is resorbable. The mechanism of osteoinduction by calcium phosphate biomaterials involves the release of ions and the formation of a bone-like apatite layer that absorbs osteogenic proteins. Surface topography and micropores also play a role in bone regeneration.

Bioactive glasses, such as 45S5 (Bioglass), are composed of calcium-containing silicates. They promote accelerated bone regeneration and bond with bone through interface reactions that form a bone-like apatite layer. The release of ions from bioactive glasses, such as Ca^{2+} , PO_4^{3-} , and Si^{4+} , contributes to their osteogenic potential. Researchers are exploring the use of doped bioactive glasses and mesoporous bioactive glasses loaded with drugs or biomolecules to enhance their functionality. Bioceramics offer promising options for bone-tissue engineering due to their bioactivity, ability to bond with bone, and potential for controlled release of ions and drugs. They provide a scaffold for bone regeneration and can be tailored to specific applications in bone-tissue engineering. Crosslinking is important for stability and gel formation in hydrogel systems. Covalent and non-covalent crosslinking strategies offer different advantages, and their

choice affects degradation behavior. Crosslinking agents must be biocompatible and extractable before exposure to living cells or tissue. Bioceramics offer promising possibilities for bone-tissue engineering due to their unique properties, but they also come with some significant disadvantages. One major drawback of bioceramics is their inherent brittleness and susceptibility to fractures. Ceramics, glasses, and glass ceramics can be fragile and may not withstand the mechanical stresses experienced in load-bearing applications, leading to potential implant failure or complications during the healing process.

Another disadvantage of bioceramics is related to their limited ability to undergo resorption or remodeling. While some bioactive ceramics, such as hydroxyapatite, exhibit slow degradation, they are not fully resorbable. In contrast, inert bioceramics cannot form a direct bond with surrounding bone tissue, which can hinder the integration of the implant with the natural bone and potentially lead to long-term stability issues. Furthermore, the release of biologically active ions from bioceramics may have both positive and negative effects. While the controlled release of ions like calcium, phosphate, and silicon can promote osteogenesis and bone regeneration, excessive or uncontrolled ion release may lead to adverse reactions or interfere with the natural cellular processes.

Additionally, the manufacturing of bioceramics can be complex and costly, especially when tailored to specific applications in bone-tissue engineering. The process of synthesizing bioceramics with precise compositions and properties requires specialized expertise and equipment, which can limit their widespread use and accessibility.

Moreover, the biocompatibility of certain bioceramics needs to be thoroughly evaluated to ensure they do not elicit harmful immune responses or toxicity when in contact with living tissues. While many bioceramics have demonstrated biocompatibility, some variations or modifications may introduce unforeseen risks.

2.2.1.3. Metals

Metals, such as titanium and magnesium, are used in bone-tissue engineering for their high mechanical performance. Metallic implants are highly valued in bone-tissue engineering due to their superior mechanical performance. Titanium and its alloys, along with magnesium, are the most commonly used metals for these applications. Titanium surfaces can be chemically modified to promote osteointegration, making them bioactive and encouraging a favorable host immune response. However, titanium alone is not biodegradable, making it less than ideal for long-term use in bone-tissue engineering. Pure titanium materials often require surgical removal after serving their purpose of supporting bone regeneration.⁴

To address the drawbacks of titanium, titanium-containing materials with reduced stress shielding risk and bone-tissue infiltration capabilities have been developed using titanium fibers or meshes. However, concerns about wear-induced particulate debris have raised questions about potential inflammatory reactions caused by contact with bone.

On the other hand, biomaterials made of magnesium offer a promising alternative. Magnesium alloys are amenable to biodegradation by corrosion, making them more suitable for bone-tissue engineering. Clinical trials have shown that magnesium alloys undergo complete degradation and are replaced by newly formed bone tissue within a year of implantation. While rapid degradation can have adverse effects and should be carefully managed, the process increases local pH, facilitating the precipitation of calcium phosphate at the implant site. Additionally, Mg^{2+} ions and iron nanoparticles promote the upregulation of osteogenic growth factors produced in MSCs (mesenchymal stem cells). Furthermore, the incorporation of Ag^{+} ions into biomaterials provides antibacterial properties, which is particularly beneficial in tissue-engineering strategies to prevent infections.

Titanium provides excellent mechanical performance but lacks biodegradability, necessitating removal after bone regeneration. On the other hand, magnesium alloys can degrade and be replaced by bone tissue, but careful management is required to avoid adverse effects from rapid degradation.

2.2.2. Fabrication Technique

The fabrication technology used in biomaterials plays a crucial role in determining their properties. Different fabrication techniques can influence the morphological, physicochemical, mechanical, and biological characteristics of the biomaterials. The choice of fabrication technique depends on the desired form and architecture of the biomaterial, as well as the

processing requirements of the material.

Various fabrication methods can be employed, depending on the application and strategy. These include the formation of microparticles or nanoparticles, fabrication of fibers, coatings, films, and three-dimensional (3D) constructs. Emulsion-based techniques, such as oil-in-water and water-in-oil-in-water emulsions, are commonly used for particle formation. Coatings and films are typically fabricated using methods like dip coating, physical vapor deposition, chemical vapor deposition, and layer-by-layer deposition. For 3D constructs in bone-tissue engineering, conventional techniques like solvent casting, particulate leaching, foam replica, gas foaming, and thermally induced phase separation have been widely used. However, these techniques have limitations, such as poor scalability, and may not offer precise control over scaffold microstructure. Two advanced fabrication methods that are increasingly preferred for bone-tissue engineering are 3D printing and electric-field-assisted techniques. 3D printing allows for precise control over the architecture and microstructure of the scaffolds, enabling the fabrication of complex geometries and patient-specific designs. Electric-field-assisted techniques, such as electrospinning and electrospraying, utilize electric fields to fabricate fibers or droplets, respectively, with high precision and control. These advanced fabrication methods offer scalability, improved control over scaffold microstructure, and the ability to create customized designs for bone-tissue engineering applications. They are at the forefront of biomaterials design and hold significant promise for the development of functional and patient-specific scaffolds.

- What is 3D printing technology

3D printing, also known as additive manufacturing, has become a powerful tool for fabricating bone-tissue engineering scaffolds and conducting bioprinting. With the expiration of important patents, 3D printing has become more accessible and cost-effective. It allows designers to precisely dictate the structure and composition of scaffolds, resulting in superior pore interconnectivity, mechanical strength, and the ability to incorporate complex topological features. There are various ways to 3D print like Fused Deposition Modeling (FDM) or Fused Filament Fabrication (FFF), Stereolithography (SLA), Selective Laser Sintering (SLS), Selective Laser Melting (SLM), Digital Light Processing (DLP), Electron Beam Melting (EBM), Binder Jetting, Material Jetting, Direct Metal Deposition (DED), Laminated Object Manufacturing (LOM), Continuous Liquid Interface Production (CLIP), Powder Bed Fusion

(PBF), Biofabrication/Bioprinting

Different forms of 3D printing used in bone-tissue engineering include extrusion, stereolithography, selective laser sintering, and inkjet printing. Extrusion printing involves depositing filaments in a distinct pattern layer by layer, using pneumatic pressure or mechanical plunger extrusion. Thermoplastic materials can also be used in a continuous filament form, known as fused deposition modeling (FDM). Ceramic or glass-based slurries can be directly extruded to fabricate green bodies for bioceramic scaffolds.

- Advantages

3D printing offers numerous advantages in tissue engineering, revolutionizing the field and providing new opportunities for regenerative medicine. Some of the key advantages of 3D printing in tissue engineering include:

1. **Customization and Patient-Specific Solutions:** 3D printing enables the creation of highly customized tissue constructs tailored to individual patients. By using patient-specific data, such as medical imaging, researchers can design and fabricate tissue scaffolds that precisely match the patient's anatomy and specific needs. This customization enhances tissue compatibility and promotes successful integration with the host tissue.
2. **Complex Geometry and Porosity:** 3D printing allows the fabrication of complex three-dimensional structures with intricate geometries and porosities. This capability is crucial for mimicking the natural architecture of tissues, promoting cell infiltration, nutrient diffusion, and waste removal. The controlled pore size and distribution can also influence cell behavior and tissue regeneration.
3. **Rapid Prototyping and Iterative Design:** 3D printing accelerates the tissue engineering process by enabling rapid prototyping and iterative design. Researchers can quickly create and test different scaffold designs, optimizing their properties and performance based on in vitro and in

vivo experiments.

4. Multi-material Printing: Advanced 3D printing techniques enable the use of multiple materials within the same scaffold. This allows for the incorporation of bioactive agents, growth factors, and different types of cells, facilitating the development of biomimetic and functional tissue constructs.

5. Biodegradable Materials: 3D printing with biodegradable materials is particularly advantageous in tissue engineering. Biodegradable scaffolds gradually degrade over time, providing temporary support for tissue regeneration while gradually being replaced by the new tissue. This feature eliminates the need for scaffold removal surgeries.

6. Vascularization: Vascularization is a critical challenge in tissue engineering, as it ensures sufficient nutrient supply and oxygen delivery to support the growing cells. 3D printing allows for the creation of intricate vascular networks within the scaffold, facilitating the formation of functional blood vessels and improving the survival and integration of the engineered tissue.

7. Scale-up and Mass Customization: 3D printing offers scalability, making it feasible to produce tissue constructs at various sizes, from small laboratory samples to larger implantable tissues. This potential for mass customization is essential for addressing individual patient needs and clinical requirements.

8. Reduced Material Waste: 3D printing is an additive manufacturing process, meaning materials are deposited layer by layer, resulting in minimal waste generation. This contrasts with traditional subtractive manufacturing methods, which often produce significant material waste.

9. Organ-on-a-Chip technology: 3D printing can be used to create microfluidic devices better known as “organ on chips”, which mimics the structure and function of organs. These devices offer a cost-effective and scalable way to study organ-level responses to drugs and disease

without using live animals.

Stereolithography

Stereolithography offers higher resolution and utilizes transparent, low-viscosity liquid resins that can be solidified at specific locations using light patterns. Projection-based stereolithography enables the fabrication of hydrogel-based constructs with internal channels or complex vascular networks. Volumetric printing techniques can produce anatomically shaped constructs that mimic the trabecular architecture of bone. Selective laser sintering involves selectively fusing particulate materials arranged as a powder bed using a laser beam. This technique is suitable for metallic, polymeric, and ceramic biomaterial constructs and does not require the inclusion of temporary support structures. Inkjet printing enables the localized deposition of individual material droplets, stabilized by subsequent UV curing. Two-photon polymerization allows for the fabrication of features with sizes smaller than 1 μm , mimicking the porous architecture of trabecular bone. Post-processing techniques can be applied to improve the resolution and properties of printed constructs. Hydrothermal processing, combinations of different printing methods, or post-printing processing can enhance the freedom of fabrication choices and modulate scaffold-material properties and interactions with cells.

Methods

3.1. CAD modelling for scaffold designing

The process of making 3D printed Scaffolds involves computer-assisted design (CAD) software to generate a precise 3D model of a specific bone defect directly from clinical imaging data. Most current bone implants produced through 3D printing are made from permanent, non-degrading materials designed for replacement rather than tissue regeneration. These implants often utilize mirroring techniques, where the opposite anatomical region, containing healthy bone tissue, is scanned, and a mirrored image is used as the CAD template, particularly for large bone defects lacking tissue entirely. Despite this, a few case reports have demonstrated the clinical implementation of CAD-derived degradable scaffolds for bone regeneration. For instance, in a pediatric patient, a cell-seeded PCL (polycaprolactone) scaffold was successfully used to reconstruct a bone defect in the upper jaw. A CAD model of the bone defect, derived from computed tomography scans, is employed in the fabrication of the scaffold using

extrusion-based 3D printing technology. There are many different types of CAD software available like AutoCAD, SolidWorks, CATIA, Siemens NX (formerly Unigraphics NX) PTC Creo (formerly Pro/ENGINEER), Autodesk Inventor, Rhinoceros (Rhino), Blender, SketchUp, FreeCAD, Fusion 360, Onshape, ZBrush. The software I used for the 3D model of the scaffolds is SolidWorks. The shapes used for designing the scaffolds are very elementary like cuboidal-octagonal grids. These designs do not require any supporting structure. The design started from selecting elementary shapes like rectangle and octagon from the preoccupied option or can be designed by aligning lines in particular interior angles. Once the 2D shape is complete, options like extrude or loft tools can be used in order to turn it into a 3D structure. The reason for designing the structure as grids is because it offers a degree of porosity. This porosity is essential for the exchange of nutrition, oxygen and waste products between the scaffold and the growing cells. Grid-based scaffolds can exhibit excellent mechanical properties due to their lattice-like arrangement. The grid design allows for a good balance between structural integrity and flexibility, providing the necessary support for cell growth while allowing the scaffold to withstand external forces. The length breadth and height of the scaffolds are $5 \times 5 \times 0.5$ mm respectively.

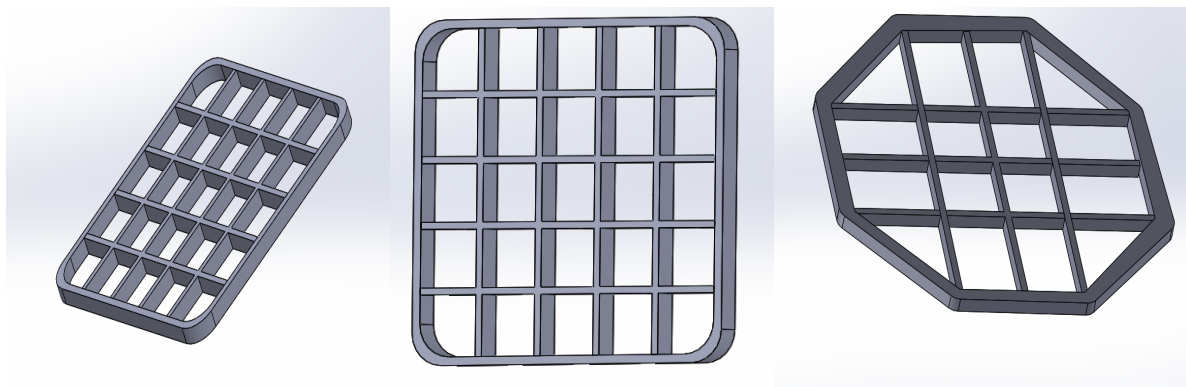


Fig.2. Square and Octagonal Scaffolds structure

Slicer

Slicing is an important part of 3D printing, by means of slicing the 3D printing of scaffolds are executed in layers. The Slicer converts the stl files into G Code which is comprehensible for the Printer. Also it is an important part where we adjust the material's infill density as well as its printing speed and the consecutive number of layers it is set to print. The 3D printer we have used is Original Prusa i3 MK3 with slicing software Prusa Slicer.

3.2. 3D Printing of Polymeric scaffold -

We used the Original Prusa i3 MK3 for slicing our stl files. Once the files are imported in stl format we change the size of the structure into a desirable dimension as well as change the constraints like speed and infill density. It is advisable to check whether the structure requires an additional supporting structure. Now, we click on create G code and import the G code file with the help of any memory device. We need to connect the memory device to the Original Prusa i3 MK3. Preheat the 3D printer in order to clean the nob if any waste is remaining inside the nob. Maintain the nob and bed temperature and select the file from the memory device inserted. Print the acquired file and maintain the temperature as well as the nob speed. At last the additional supporting structure from the desirable scaffolds needs to be removed.

4. Results-

The 3D printing of structures using modified STL files on the Original Prusa i3 MK3 3D printer occurred successfully. The STL files were accurately sliced and generated the appropriate G-code. The 3D printer performed well, maintaining the necessary temperatures and nozzle speed during printing, resulting in print scaffolds. The experiment validated the effectiveness of using the Original Prusa i3 MK3 3D printer for printing structures with modified STL files. The ability to adjust the size and constraints of the objects allowed for greater customization and flexibility in the 3D printing process.

During the 3D printing process of the scaffolds using the Original Prusa i3 MK3 printer, a noise error was encountered due to the limited resolution of the printer's nozzle. The small size of the scaffolds, specifically measuring $5 \times 5 \times 0.5$ mm, proved to be inadequate for the printer's capabilities. The nozzle resolution of the Original Prusa i3 MK3 is typically around 0.4 mm, which made it challenging to accurately reproduce intricate details in such tiny structures. As a

result, the noise error manifested as inconsistencies and imperfections in the printed scaffolds, affecting their overall quality and structural integrity.

To address this issue and improve the printing of small-sized scaffolds, several steps can be taken. Firstly, it is essential to increase the size of the scaffolds slightly, allowing for better compatibility with the printer's nozzle resolution. By enlarging the structures just enough to align with the printer's capabilities, the risk of noise errors can be significantly reduced.

Additionally, optimizing the slicer settings is crucial for achieving finer details. Adjusting the layer height and nozzle temperature can help in refining the printing process, enabling better precision for small objects. Lowering the layer height can improve the vertical resolution, while adjusting the nozzle temperature can aid in better material flow and adhesion.

Furthermore, exploring the option of using a smaller nozzle size, if available for the Original Prusa i3 MK3, can enhance the printer's ability to handle intricate details in smaller structures. A reduced nozzle size, such as 0.2 mm, would enable higher resolution printing and potentially mitigate noise errors. To address the noise error arising from printing small scaffolds with the Original Prusa i3 MK3, it is advisable to resize the structures to better suit the printer's nozzle resolution. Fine-tuning slicer settings and considering a smaller nozzle size can also contribute to improved print quality and increased accuracy, ensuring successful 3D printing of intricate and delicate objects.

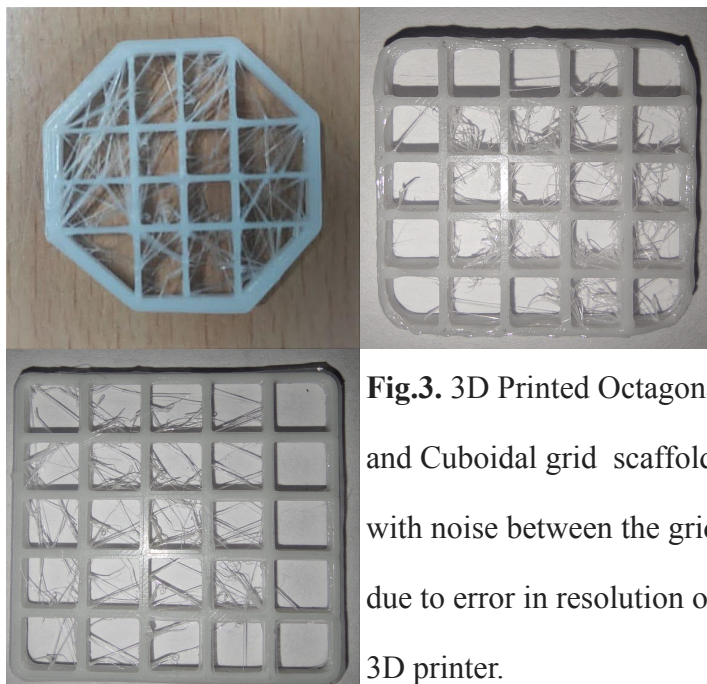


Fig.3. 3D Printed Octagonal and Cuboidal grid scaffolds with noise between the grids due to error in resolution of 3D printer.

5. Conclusion -

Despite significant progress in the development of biomaterials for bone tissue engineering, there are still challenges and unmet needs that hinder scientific progress and clinical translation. One major obstacle is the limited understanding of the mechanisms of action and cellular response to biomaterials, which necessitates systematic studies. However, it is difficult to decouple different material properties to identify their specific effects on cellular response. Additionally, conflicting requirements often arise, such as the need for both mechanical stability and rapid degradability, requiring the use of novel materials and complex design strategies.

Another challenge is the discrepancy between the viscoelastic nature of body tissues and the focus on fully elastic materials in bone tissue engineering. Viscoelastic properties, such as stress relaxation, influence cell behaviors and the remodeling process. Hydrogels with tunable stress-relaxation behavior show promise for directing bone tissue regeneration, but their mechanical weakness can be improved by incorporating nanomaterials or interpenetrating polymer networks. Tissue regeneration is a dynamic process involving interactions between cells and the surrounding matrix. Biomaterials that can adapt to local cellular changes and respond to biological signals are desired. Adaptive and responsive properties can be achieved through bottom-up strategies, leveraging the chemistry, structure, size, and shape of building blocks.

Bone tissue formation occurs at multiple length scales, and controlling biomaterial features at different scales requires the combination of multiple fabrication techniques. Surface topography, including features comparable to or smaller than cells, can influence cell differentiation. 3D printing methods enable the fabrication of biomaterials and tissue-like constructs with complex geometries and spatially distinct functions. Microfluidics-based fabrication methods offer powerful tools for dictating structure at the microscale and nanoscale levels.

Scalability is crucial for the clinical and commercial success of bone tissue engineering strategies. Both material synthesis and the fabrication of large constructs need to be considered. The clinical handling and delivery method of biomaterials also require special consideration. Injectable biomaterials offer minimally invasive application, but maintaining their integrity in the defect site can be challenging. Prefabricated foldable 3D constructs that can be injected and re-expand upon implantation show promise. Biomaterials with self-healing properties can address mechanical damage, and modular biomaterial systems are needed to enable precision medicine.

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